ELECTRIC VEHICLE MARKET PERFORMANCE & INSIGHTS

DATA ANALYTICS WITH TABLEAU CAPSTONE PROJECT

Final Project Report

Project	Electric Vehicle Market Performance & Insights
Tableau Public Workbook	Electric Vehicle Market & Technology Analysis
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Enrollment Number	EN25CS3011105
Institution	Medicaps University, Indore
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Project Website	https://en25cs3011105-spec.github.io/Electric-Vehicle-Market-Performance-and-Insights/

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This report documents the project scope, CSV-based data preparation, Tableau analysis, dashboard, current story completion status, testing evidence and future enhancements.

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Project status note: The Tableau workbook and project website are published. The Tableau Story has only Story Point 1 completed. Story Points 2–4 are proposed future work and are not represented as finished.

1. INTRODUCTION

1.1 Project Overview

Electric Vehicle Market Performance & Insights is a data analytics capstone project that uses a supplied electric-vehicle market dataset and Tableau Public to explore sales performance, vehicle categories, manufacturers, battery technologies, range, price, customer ratings and manufacturing geography. The project presents the analysis through interactive worksheets, a dashboard and a published project website.

The Tableau workbook is titled **Electric Vehicle Market & Technology Analysis**. The dashboard combines summary measures and visual comparisons so that a viewer can explore patterns across vehicle model year, category, battery type and manufacturing country.

1.2 Purpose

The purpose is to convert a multi-field EV dataset into accessible visual insights that can support market understanding and strategic discussion. The analysis is intended to help a reader compare vehicle segments, understand battery chemistry adoption and range, examine price versus range, and view the geographic distribution of manufacturing and sales.

The project is descriptive and exploratory. It does not claim to forecast future sales, establish causal relationships or provide verified investment recommendations.

2. IDEATION PHASE

2.1 Problem Statement

Electric-vehicle market data contains multiple dimensions and performance measures. Reviewing the raw data alone makes it difficult to compare category sales, manufacturer contribution, battery technology, vehicle range, price and manufacturing footprint in one place. The project addresses this information-discovery problem by organizing the dataset into interactive Tableau visualizations and a web-accessible presentation.

2.2 Empathy Map Canvas

Persona: Emily Johnson is a hypothetical customer persona used to frame the dashboard audience. She is not a real customer and no direct interview or user research with her is claimed.

Empathy dimension	Hypothetical persona perspective
Says	“I want to compare EV categories, battery options, range and price before understanding the market.”
Thinks	“Which battery technologies and vehicle segments appear strongest in this dataset?”
Does	Uses dashboard filters and selects charts to compare categories, battery types, manufacturers and countries.
Feels	Wants the information to be clear, comparable and easy to navigate without manually scanning raw rows.

This empathy map is a design assumption for the intended audience, not evidence from a real interview.

2.3 Brainstorming

- Create an executive summary of key market measures.
- Compare annual sales units across vehicle categories.
- Show manufacturer contribution using a treemap.
- Compare median vehicle range across battery types and battery adoption share.
- Map manufacturing countries to reveal geographic patterns.
- Use a scatter plot to examine average price against average range.
- Compare customer rating and battery capacity by category.
- Build a country-versus-battery matrix to inspect distribution.
- Provide filters and dashboard actions so users can explore subsets of the data.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User activity	Dashboard support
Discover	Opens the project website and dashboard.	Landing page with project title and links to the Tableau visualization.
Orient	Reviews headline KPIs and available filters.	Executive summary and filters for model year, category, battery type and manufacturing country.
Explore	Compares sales, manufacturers, batteries, range, price and ratings.	Nine worksheets and interactive dashboard actions.
Interpret	Looks for patterns and outliers in the selected views.	Labels, axes, color encoding, map and matrix views.
Conclude	Forms a descriptive understanding of the dataset.	Story Point 1 provides a guided overview of category sales and manufacturer share.

3.2 Solution Requirement

- Input: supplied CSV training dataset with vehicle, manufacturer, category, battery, commercial, geographic and sustainability fields.
- Data preparation: remove nulls and duplicates and correct data types as reported by the project owner.
- Analysis: use Tableau aggregations, distinct counts, averages, medians and percent-of-total calculations.
- Visualization: build charts and a dashboard with filters and cross-filter actions.
- Publication: publish the workbook to Tableau Public and provide a GitHub Pages website for project access.
- Documentation: accurately state scope, assumptions, completion status and test limitations.

3.3 Data Flow Diagram

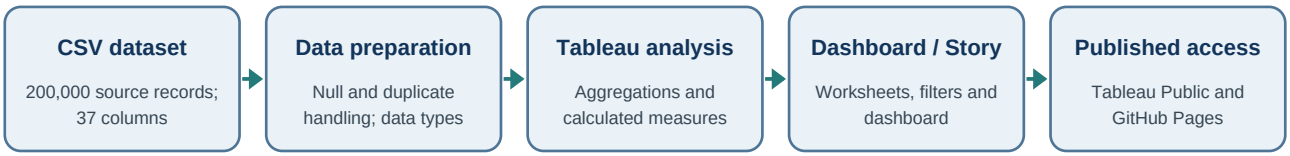

Figure 1. High-level data flow. The supplied test CSV was used only as a resource and was not connected to the Tableau workbook.

3.4 Technology Stack

Layer	Technology / resource	Role
Data source	ev_market_train.csv	Training dataset used for the Tableau analysis.
Data preparation	CSV data preparation	Null/duplicate handling and data-type correction, as reported.
Analytics and visualization	Tableau Public	Nine worksheets, one interactive dashboard and a Story.
Website	HTML, CSS and JavaScript	Public-facing project website and Tableau dashboard embed/link.
Hosting / version control	GitHub repository and GitHub Pages	Stores website files and serves the public website.
Currency convention	USD ($)	Dashboard price measures are expressed in USD.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The selected solution fits the problem because Tableau can combine categorical comparisons, aggregated measures, geographic views and interactive filters in one published interface. A static spreadsheet would require users to manually construct comparisons; the dashboard provides a more direct exploratory workflow.

4.2 Proposed Solution

The proposed solution is an interactive Tableau analysis supported by a simple public website. The website introduces the project and directs visitors to the embedded or linked Tableau Public workbook. The workbook includes nine worksheets and a dashboard. The Story is a separate guided narrative and is currently only partially completed.

4.3 Solution Architecture

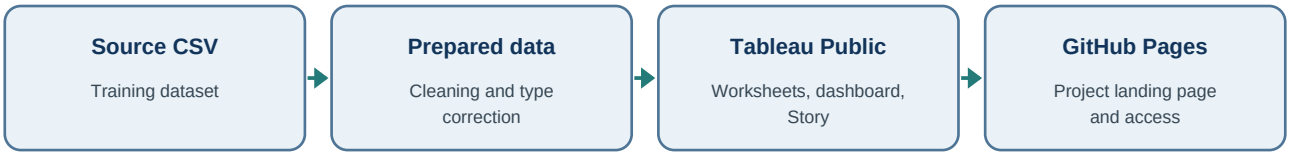

Figure 2. Solution architecture. Snowflake is not part of the implemented technology stack.

The project uses a file-to-visualization workflow: the CSV dataset is prepared, connected to Tableau Public, analyzed through worksheets, assembled into a dashboard and published. The GitHub Pages site acts as a public project entry point. There is no Snowflake database, SQL warehouse, API service or live data pipeline in the confirmed implementation.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Period	Planned / completed activity
1–3 October 2026	Understand the project brief, identify the EV market analysis objectives and review the supplied dataset.
4–7 October 2026	Inspect the CSV fields, identify dimensions and measures, and plan the analytical questions.
8–12 October 2026	Prepare the data by handling nulls and duplicates and correcting data types.
13–17 October 2026	Build and refine the Tableau worksheets for category sales, manufacturer share, battery range/adoption, geography and price-range analysis.
18–21 October 2026	Assemble the dashboard, arrange views and configure filters and dashboard actions.
22–24 October 2026	Create the Tableau Story structure and complete Story Point 1: Global EV Market & OEM Overview.
25–27 October 2026	Publish the workbook and website, review the outputs and prepare project documentation.
28 October 2026	Final submission date.

The schedule records the project timeline supplied for documentation. It does not imply that the planned Story Points 2–4 were completed.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The project was reviewed for successful data loading and rendering of the Tableau worksheets, dashboard and published website. No controlled benchmark was supplied for page-load time, filter response time, workbook query execution time or browser performance. Therefore, numerical speed claims are not made.

Test area	Observed / documented result
Source file inspection	Uploaded training CSV: 200,000 rows × 37 columns; 46,478,548 bytes (approximately 44.33 MiB).
Source data quality inspection	Uploaded source file contains 18,996 null cells across 18,251 rows, 1 exact duplicate row and 1 repeated vehicle_id.
Cleaning process	Project owner confirms nulls and duplicates were removed and data types corrected. The uploaded file appears to be the source CSV, not a verified cleaned export; post-cleaning counts were not provided.
Data source scope	ev_market_train.csv was used for the Tableau workbook. ev_market_test.csv was supplied as a resource only and was not connected to the workbook.
Dashboard filters	model_year, category, battery_type and manufacturing_country.
Dashboard actions	“Use as Filter” actions are enabled for interactive exploration.
Story test	Story Point 1 is completed. Story Points 2–4 remain planned / incomplete.
Snowflake test	Not applicable. Snowflake was not used in this project.

Source-file inspection and cleaning evidence must not be confused: the null and duplicate counts above describe the uploaded source CSV. The project owner reports that cleaning was performed, but a cleaned export or before/after cleaning evidence was not supplied. The figures must not be presented as the final cleaned-data counts.

No measured dashboard response times, load-time results, Snowflake query history or SQL performance data are available. These tests are not claimed as completed.

7. RESULTS

7.1 Output Screenshots

The following output views are documented from the Tableau workbook and screenshots previously supplied for this project. The actual screenshots should be attached as evidence in the SkillWallet portal or inserted into the editable report if required.

Output	What it presents
Executive Summary / KPI view	Annual sales units, average price in USD, average range and average customer rating.
Category Sales	Horizontal bar chart ranking EV categories by annual sales units.
Manufacturer Share (Treemap)	Treemap showing annual sales distribution across manufacturers.
Battery Performance	Median range comparison across LFP, NCA, NMC, Sodium-Ion and Solid-State batteries.
Battery Tech Share	Pie chart showing battery-type share.
Global Manufacturing Map	Geographic view of annual sales by manufacturing country.
Price vs Range	Scatter plot comparing average range and average price by vehicle category, with a linear trend line.
Rating vs Battery / Category	Combined view comparing customer rating and battery capacity by category.
Country vs Battery Matrix	Highlight table comparing vehicle counts across battery types and manufacturing countries.
Story Point 1	Global EV Market & OEM Overview, using Category Sales and Manufacturer Share.

Selected values visible in the supplied screenshots include executive summary annual sales of approximately 1,550.70M units, average price approximately USD 64.92K, average range 325 km and average customer rating 4. Battery Tech Share labels show NMC 33.80%, LFP 31.99%, NCA 17.23% and Sodium-Ion 9.88%; the remaining share is Solid-State. Battery Performance labels show Sodium-Ion at 341 km median range, with the other displayed battery types around 337–338 km. These are screenshot-visible, rounded values and may vary with filters.

The supplied screenshot evidence does not show every dashboard element at full resolution. The final report therefore describes the outputs rather than embedding reconstructed or fabricated screenshots.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Combines multiple EV market dimensions in a single interactive analysis.

- Allows comparison of category sales, manufacturers and battery technologies.
- Uses different visual forms suited to the data: bars, treemap, pie, map, scatter plot and highlight table.
- Provides filters and dashboard actions for user-driven exploration.
- Published Tableau workbook and GitHub Pages site make the project accessible online.
- Uses a clearly stated USD currency convention for price measures.

Disadvantages and limitations

- The analysis is descriptive and depends on the quality and coverage of the supplied dataset.
- The uploaded source CSV contains null values and a duplicate row; the cleaned output was not provided for independent verification.
- No real-time EV market feed or automated refresh pipeline is implemented.
- No controlled performance benchmarks were recorded.
- The Tableau Story is incomplete beyond Story Point 1.
- Emily Johnson is a hypothetical persona, not a validated customer research participant.

9. CONCLUSION

The project demonstrates how a structured EV dataset can be explored through Tableau Public to understand category sales, manufacturer distribution, battery technology, vehicle range, pricing, customer ratings and manufacturing geography. The work includes nine worksheets, an interactive dashboard and a public-facing website. The Story has begun with the global EV market and OEM overview. The project provides a usable descriptive analytics deliverable while clearly identifying evidence gaps and unfinished narrative work.

10. FUTURE SCOPE

- Complete Story Point 2 on battery chemistry, adoption and range, Story Point 3 on pricing and satisfaction, and Story Point 4 on manufacturing footprint and strategic recommendations.
- Verify the cleaned CSV by exporting it and calculating post-cleaning row, null and duplicate counts.
- Add dashboard documentation for filter behavior and user instructions.
- Conduct repeatable performance tests for workbook opening, dashboard rendering and filter response time, recording the environment and test method.
- Add additional validated data sources and a scheduled refresh process if the project scope permits.
- Improve dashboard accessibility with consistent labels, explanatory tooltips and clearer units.
- Document the limitations of the dataset and avoid interpreting correlations as causal effects.

11. APPENDIX

A. Dataset details

The uploaded file inspected for this report is **ev_market_train(1).csv**. It contains 200,000 records and 37 columns and has a file size of 46,478,548 bytes (44.33 MiB). The training file is the dataset used for the Tableau project. The separate test CSV was not connected to the workbook.

Identifiers and market segments

vehicle_id, model_name, manufacturer, model_year, manufacturing_country, market_region, category

Battery and vehicle specifications

battery_capacity_kwh, battery_type, battery_cycle_life, battery_degradation_pct_5yr, range_km, efficiency_wh_per_km, motor_power_kw, torque_nm, num_motors, drivetrain, top_speed_kmph, acceleration_0_100_sec, max_dc_charging_kw, charging_dc_fast_time_min, charging_ac_time_hr, num_charging_ports, num_seats, weight_kg, cargo_capacity_liters

Commercial and ownership measures

price_usd, insurance_cost_usd_yearly, warranty_years, resale_value_pct_3yr, safety_rating_stars, autonomous_level, infotainment_screen_inch, co2_saved_tons_per_year

Sustainability, customer and use-case fields

annual_sales_units, customer_rating, recommended_use_case

B. Data preparation note

Source file inspection results: 18,996 null cells, 18,251 rows containing at least one null, 1 exact duplicate full row and 1 repeated vehicle_id. The project owner confirms that nulls and duplicates were removed and data types corrected during preparation. Because the uploaded file is the source file and no cleaned export was supplied, post-cleaning totals cannot be verified.

C. Tableau calculations and measures

- SUM(annual_sales_units)
- COUNTD(vehicle_id)
- AVG(range_km)
- AVG(price_usd)
- AVG(customer_rating)
- MEDIAN(range_km)
- Percent of Total calculation for share analysis
- AVG(battery_capacity_kwh) for battery-related comparison

D. Source code

The Tableau workbook contains the analytical implementation. No separate source-code archive was provided for this report. The website repository contains the website implementation. Do not claim a separate Python/SQL application or Snowflake pipeline unless one is actually part of the submitted repository.

E. Dataset link

Dataset source: the supplied SkillWallet training resource. A public dataset download URL was not provided in the project information.

F. GitHub and project demo links

GitHub Pages website: https://en25cs3011105-spec.github.io/Electric-Vehicle-Market-Performance-and-Insights/

Tableau Public workbook: https://public.tableau.com/views/ElectricVechileMarketTechnologyAnalysis/ElectricVehicleMarketPerformanceInsights?:showVizHome=no

G. Project ownership and persona note

The project was completed individually by Yuvraj Shrivastava (EN25CS3011105), Medicaps University, Indore. Emily Johnson is a hypothetical customer persona used for design framing only.